Zhou et al. BMC Plant Biology (2025) 25:35
https://doi.org/10.1186/s12870-024-05788-w

BMC Plant Biology

RESEARCH

Open Access

Genome-wide identification of CAMTA gene family in teak (*Tectona grandis*) and functional characterization of *TgCAMTA1* and *TgCAMTA3* in cold tolerance

Wenlong Zhou[1], Jian Du[2], Runjie Jiao[1], Xianbang Wang[1], Tiansong Fang[3*] and Guihua Huang[1*]

Abstract

Background Calmodulin-binding transcription activator (CAMTA) proteins play significant roles in signal transduction, growth and development, as well as abiotic stress responses, in plants. Understanding their involvement in the low-temperature stress response of teak is vital for revealing cold resistance mechanisms.

Results Through bioinformatics analysis, the CAMTA gene family in teak was examined, and six *CAMTA* genes were identified in teak. The encoded proteins were predicted to be located in the nucleus and exhibited hydrophilic properties, with molecular weights ranging from 103.4 to 123.3 kDa and isoelectric points ranging from 5.49 to 7.55. On the basis of protein sequence homology, the CAMTA family could be divided into three subgroups. Domain and 3D structure analyses demonstrated that all the TgCAMTA proteins contained the typical CAMTA domain with the CaMBD binding domain, which was exposed on the surface. Expression analysis of different tissues revealed the expression of *TgCAMTA* genes in teak roots, stems, leaves, flowers, fruits, and branches. Furthermore, the promoter region contained various cis-acting elements related to light, hormone, and abiotic stress responses. After low-temperature stress treatment, different expression patterns of *TgCAMTAs* were observed in teak roots, stems, and leaves, with *TgCAMTA1* showing the highest expression level in leaves compared with stems. Transgenic lines carrying the *TgCAMTA1/3 promoter::GUS* construct cold stress induction of *TgCAMTA1/3* genes revealed the presence of multiple low-temperature responsive cis-acting elements in the *TgCAMTA1/3* promoter region. Subcellular localization analysis indicated that these genes were functional predominantly in the nucleus. Compared with wild-type Arabidopsis, *TgCAMTA1/3*-overexpressing Arabidopsis presented increased tolerance to freezing stress, with increased expression of *AtCOR* genes. Moreover, under low-temperature conditions, *TgCAMTA3*-overexpressing Arabidopsis presented significantly elevated expression levels of genes related to the CBF signaling pathway, including *AtCBF1/2/3*.

Conclusions Our findings add significantly to the existing knowledge regarding cold stress tolerance and help elucidate cold response mechanisms in teak.

*Correspondence:
Tiansong Fang
fts1973@126.com
Guihua Huang
ritfhuanggh@caf.ac.cn

Full list of author information is available at the end of the article

Keywords *Tectona grandis*, CAMTA transcription factor, Bioinformatics, Low-temperature stress, GUS, Transgenic lines

Background

Ca^{2+} is a widely present intracellular signal in all eukaryotic cells, and both biotic and abiotic stresses can cause changes in the intracellular Ca^{2+} concentration [1, 2]. In plants, there are three main classes of Ca^{2+} sensor proteins that perceive and transduce Ca^{2+} signals, namely, calmodulins (CaMs) and calmodulin-like proteins (CMLs), calcium-dependent protein kinases (CDPKs), and calcineurin B-like proteins (CBLs) [3]. Most of the functions of CaMs/CMLs are mediated through their interaction with calmodulin-binding proteins (CaMBPs), which play crucial roles in plant physiological and biochemical regulation processes [4, 5]. More than 90 types of CaMBPs have been reported, including CAMTA, bZIP, WRKY, MYB, and NAC family proteins [4, 6, 7]. Among them, calmodulin-binding transcription activator (CAMTA) proteins are the most extensively studied CaM-binding transcription factor family [8, 9].

CAMTA proteins contains highly conserved functional domains, including a specific DNA-binding CG-1 domain at the N-terminus, a TIG domain involved in nonspecific DNA binding, an ankyrin repeat domain, a Ca^{2+}-dependent calmodulin-binding domain (CaMBD), and a series of tandem repeats of the IQ motif (IQXXXRGXXX) that interact with CaM independently of Ca^{2+} [10, 11]. CAMTA transcription factors play crucial roles in plant growth, development, and response to abiotic stresses. *Arabidopsis thaliana* contains 6 members of the CAMTA family [10]. Previous studies have shown that Arabidopsis CAMTA family proteins respond to low-temperature stress through the C-REPEAT BINDING FACTOR 3 (CBF) cold signaling pathway [12, 13]. *AtCAMTA3* positively regulates the low-temperature stress response factor CBF2, whereas the double mutation of *atcamta1* and *atcamta3* reduces the tolerance of Arabidopsis to low temperatures [13]. *AtCAMTA3* and *AtCAMTA5* respond to a rapid decrease in temperature and induce the expression of *DREB1A/B/C* (also known as *CBF3/1/2*, respectively), allowing plants to efficiently control the acquisition of freezing tolerance [14]. Additionally, downstream of the CBF cold stress pathway, the late embryogenesis abundant (LEA) family proteins *AtCOR15A* and *AtCOR47* are upregulated in response to low-temperature stress in Arabidopsis [15–17]. Moreover, *AtCAMTA1* also responds to drought stress by regulating certain stress-responsive genes, including dehydration-responsive element-binding protein (RD26), early responsive to dehydration protein (ERD27), ABA-responsive element-binding protein (RAB28), lipid transfer proteins (LTPs), cold-regulated protein (COR78), and heat shock proteins (HSPs) [18]. In wheat (*Triticum aestivum*), there are 15 CAMTA family members, with each *TaCAMTA* gene responding to at least one type of abiotic stress (low temperature, drought, NaCl, or heat treatment) [19]. In poplar (*Populus trichocarpa*), there are 7 *CAMTA* family members, and the expression of *PtCAMTA4.2* in leaves significantly increases under low-temperature stress [20]. In summary, members of the CAMTA transcription factor family play crucial roles in the plant response to stress, particularly low-temperature stress.

Teak (*Tectona grandis* L. f.) belongs to the Labiatae family, genus Tectona, and is known for its high strength, durability, and resistance to termites and fungi, making it widely used for producing high-quality furniture, in shipbuilding, and as a flooring material [21]. As one of the major forest tree species worldwide, teak has been introduced in more than 50 countries in Asia, Africa, Central and South America, and Oceania and is extensively cultivated globally [22]. However, teak is sensitive to low temperatures and is grown primarily in a few tropical and subtropical regions in China, such as Yunnan, Hainan, Guangdong, and Guangxi [23]. Low temperatures and frost are the most significant factors limiting teak cultivation in specific geographic locations. Investigating the low-temperature adaptation mechanisms of teak and enhancing its stress resistance are highly important for expanding the cultivation area.

On the basis of genomic data from teak [24], we conducted a study on the phylogenetic relationships, chromosomal distribution, collinearity, conserved structures, and cis-acting elements of the *TgCAMTA* gene family in teak. We confirmed that all the TgCAMTA family proteins in teak contain typical CAMTA family domains and respond to low-temperature stress. Furthermore, heterologous expression of *TgCAMTA1* and *TgCAMTA3* in Arabidopsis enhanced plant resistance to cold. These findings suggest that the *TgCAMTA* gene family plays a crucial role in the response of teak to low-temperature stress.

Methods

Genome-wide identification of the *CAMTA* gene family and analysis of the physicochemical properties of the encoded proteins in teak

The genomic data of teak were downloaded from the DRYAD database (https://datadryad.org/stash) [24]. To identify members of the *TgCAMTA* family, six AtCAMTA protein sequences were downloaded from the Arabidopsis TAIR database (https://www.arabidopsis.org/) [25] and used as query sequences. Homology analysis was conducted in the teak genomic database using the BLASTP tool, with an expected value (E value) setting of 1e-5, yielding candidate genes of the teak CAMTA family. Simultaneously, we downloaded the Hidden Markov Model (HMM) sequences of the CAMTA family members' characteristic structural domains, CG-1 (PF03859), ANK [Ank (PF00023), Ank-2 (PF12796)], and IQ (PF00612), from the InterPro database (https://www.ebi.ac.uk/interpro/) [26]. These sequences were used to search the aforementioned teak genomic database for proteins containing distinctive structural domains via the HMMER3.0 tool [27], with the E value set to 1e-5 [28]. The candidate protein sequences obtained via the two methods were then merged, and redundant sequences were removed manually in Excel. The NCBI-CDD site (https://www.ncbi.nlm.nih.gov/Structure/cdd/cdd.shtml) was utilized for verification of the analysis of conserved structural domains [29]. Finally, protein sequences containing the typical structural domains of the family were identified as members of the teak CAMTA (TgCAMTA) family.

The physicochemical properties of TgCAMTA proteins, including the number of amino acids, molecular weight, isoelectric point, and hydrophilicity, were analyzed via the online tool ExPASy ProtParam (https://web.expasy.org/protparam/) [30]. The subcellular localization of the TgCAMTA proteins was predicted via the online platform Plant-mPLoc (http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/) [31].

Phylogenetic relationship analysis of the CAMTA family in teak

On the basis of existing reports [19, 20], the CAMTA protein sequences of *P. trichocarpa* and *T. aestivum* were downloaded from the JGI (https://phytozome-next.jgi.doe.gov/) and Ensembl Plants (https://plants.ensembl.org/index.html) databases [32, 33]. MEGA11 software [34] was used to construct a multispecies phylogenetic tree, including species such as *T. grandis*, *A. thaliana*, *P. trichocarpa*, and *T. aestivum*, via the maximum likelihood (ML) method. The bootstrap value was set to 1000.

Prediction of the TgCAMTA gene structure, motif, conserved protein domains, and tertiary structure

The teak genome file was utilized for gene structure analysis via the online Gene Structure Display Server (http://gsds.gao-lab.org/) [35]. Motif analysis was executed via the online MEME software, setting the conservation motif limit to 10. The tertiary structure of the TgCAMTA family proteins was modelled via the SWISS model online platform (https://www.swissmodel.expasy.org/interactive) [36], and models with a confidence level of 70% or higher were selected. The results were visualized via PyMOL (https://pymol.org/) [37]. NCBI-CDD (https://www.ncbi.nlm.nih.gov/cdd/) was used for conserved domain analysis of the TgCAMTA protein sequences, and the results were visualized via TBtools [38].

Analysis of *TgCAMTA* gene distribution on chromosomes

The distribution of the *TgCAMTA* genes on the teak chromosomes was analyzed via TBtools on the basis of the downloaded teak genome annotation file and genome sequence from the DRYAD database. Their location information was plotted on pseudomolecules.

Analysis of the promoter regions of *TgCAMTA* family members in teak and transcriptional data analysis of different tissue sites

The 2000 bp sequence upstream of the transcription start site of each *TgCAMTA* was extracted as the promoter region [39]. The PlantCare (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/) online tool was utilized for predicting cis-acting elements in the promoter region [40]. Visualization was performed via Adobe Illustrator (AI).

To investigate the tissue expression patterns of *TgCAMTAs*, transcriptomic data from eight different tissue samples of teak were obtained from the DRYAD database (https://datadryad.org/stash/dataset/doi:https://doi.org/10.5061/dryad.77b2422) [24] and NCBI (PRJNA287604) [41]. The identified *TgCAMTA* genes were scanned against the transcriptomic data, and the expression levels of the *TgCAMTAs* were normalized to the log2 (FPKM) value. TBtools was subsequently employed to generate a heatmap illustrating the gene expression patterns.

Plant materials, growth conditions and cold treatments

Uniformly growing clones of "7029" teak plants were selected from the nursery of the Tropical Forestry Research Institute, Chinese Academy of Forestry (23.18°N, 113.36°E). The seedlings were subjected to low-temperature treatment after 30 days of cultivation in a greenhouse at 30 °C and 50–70% humidity. The tray-grown teak seedlings were placed in a growth chamber at 4 °C under 57% humidity, a light intensity of 206 UML, and a light cycle of 16 h/8 h to induce low-temperature stress [12]. Root, stem, and leaf samples were collected at 0 h (ck), 1 h, 3 h, 6 h, 12 h, and 24 h after-treatment, immediately flash-frozen in liquid nitrogen, and then stored at -80 °C for subsequent RNA extraction.

The *A. thaliana* ecotype Columbia (Col-0) was used to express and analyze the *TgCAMTA1/3* gene and its promoter. Seeds of overexpression Arabidopsis of *TgCAMTA1* and *TgCAMTA3* were sterilized and placed in a square plate with a side length of 10 cm containing 40 mL of sterile half MS medium, 0.8% (w/v) sucrose and 0.8% (w/v) agar (pH 5.9). After 6 days of growth under normal conditions (16 h light/8 h dark, 22 °C, 80% humidity), the plate was subjected to low-temperature stress treatment. The plates were stored at 4 °C for 3 days and transferred to a growth chamber at 12 °C for 15 days (16 h light/8 h dark, 80% humidity) for chilling stress [14, 42]. The seedling in all the square plates were grown vertically for phenotypic analysis. Moreover, the length of the roots was also recorded, and the samples were photographed before or after low temperature stress. Seeds from transgenic Arabidopsis plants were sterilized and placed on 9-cm-diameter plates containing 30 mL sterile half MS medium, 0.8% (w/v) sucrose, and 0.8% (w/v) agar (pH 5.9) for germination as part of the analysis of the *TgCAMTA1/3* gene promoter. After 15 days of growth under normal conditions (16 h light/8 h dark, 22 °C, 80% humidity), the *TgCAMTA1 promoter::GUS* and *TgCAMTA3 promoter:: GUS* transgenic plants were relocated to an incubator set at 4 °C for a period of 24 h in preparation for subsequent GUS staining [42, 43]. Sampling was performed at 0 h, 1 h, 3 h, 6 h, 12 h, and 24 h to facilitate subsequent RNA extraction procedures.

RNA extraction and real-time quantitative PCR

Total RNA extraction was performed in the same manner as for RNA-seq. A HiScript® III 1st Strand cDNA Synthesis Kit (Vazyme, Nanjing, China) was used for first-strand cDNA synthesis, and approximately 1 μg of total RNA was used for reverse transcription. RT−qPCR primers (Supplementary Table 1) (Tsingke Biotech Co., Ltd., Beijing, China) were designed via Primer 3plus (https://www.primer3plus.com/), with the elongation factor *TgEf-1α* chosen as the reference gene for normalization [44]. Primer specificity was validated by RT−qPCR melting curve analysis and agarose gel electrophoresis for specific band confirmation. qPCR was performed via a Roche/Light Cycler 480II (Roche, Switzerland) detection system and a Taq Pro Universal SYBR qPCR Master Mix kit (Vazyme, Nanjing, China) following the manufacturer's instructions. A total of 2 μl of cDNA was added to the 20 μl system for RT−qPCR. The relative expression levels of the *TgCAMTAs* were calculated via the $2^{-\Delta\Delta Ct}$ method [45]. The data are presented as the means ± standard deviations (SDs), and all the experiments were conducted in triplicate.

Subcellular localization of the TgCAMTA1 and TgCAMTA3 proteins

To construct the fusion vectors for subcellular localization analysis, the full-length CDSs of TgCAMTA1/3 without a stop codon were inserted at the *KpnI*/*XhoI* sites of the pbi121 GFP vector driven by the CaMV35S promoter (Supplementary Fig. 1). *TgCAMTA1*::GFP, *TgCAMTA3*::GFP and the nuclear localization protein SV40 NLS were transformed into GV3101 [46], and the OD at a wavelength of 600 nm of the bacterial culture was adjusted to 0.6–0.8. The target gene vector strains were mixed with the nuclear localization vector SV40 NLS strains at a 1:1 ratio and injected into the subepidermal layer of the leaves of 1-month-old *Nicotiana tabacum* plants. After two days of dark culture, the subcellular localization results were observed under a ZEISS Imager D2 fluorescence microscope (Carl Zeiss, Germany). The primers used are listed in Supplementary Table 1.

Transgene analysis of *TgCAMTA1* and *TgCAMTA3* in *Arabidopsis thaliana*

To construct the overexpression vectors for *A. thaliana*, the full-length CDSs of *TgCAMTA1/3* were amplified from cDNA via primers containing *KpnI*/*BamHI* restriction sites (Supplementary Table 1). Moreover, We use Vazyme online sites for primer design (https://crm.vazyme.com/cetool/singlefragment.html). The amp lified products were inserted into the digested pCAMBIA-2301 vector via the ClonExpress II One Step Cloning Kit (Vazyme, Nanjing, China) (Supplementary Fig. 1). The resulting overexpression vector, *35 S::TgCAMTA1/3*, was subsequently transformed into Agrobacterium strain GV3101, which was then introduced into Arabidopsis via the floral dip method [47]. Positive lines of *A. thaliana* were selected on a 1/2 MS medium that contained 50 mg/mL kanamycin. The T2 generation was generated by using the antibiotics-selectable marker-sensitive strain with a segregation ratio of 3:1. T3 homozygous progeny was affirmed from a T2 generation by RT-qPCR and two positive transgenic lines with the highest expression were chosen for developing stable T3 generation. For further experiments, T3 generation of transgenic *Arabidopsis thaliana* was used.

Additionally, the 2000 bp region upstream of the start codon ATG of the full-length CDS of *TgCAMTA1* and *TgCAMTA3* was selected as the promoter region. This promoter region was amplified via specific primers to construct the pUBI10-*TgCAMTA1 promoter::GUS* and pUBI10-*TgCAMTA3 promoter::GUS* vectors (Supplementary Fig. 1), which were subsequently transformed into Arabidopsis via the aforementioned method. Every *TgCAMTA1 promoter::GUS* and *TgCAMTA3 promoter::GUS* transgenic plants generated 16 unique T3 lines, of which 2 or 3 were selected and exhibited identical staining as the rest; these were used for further investigation. All primer sequences are provided in Supplementary Table S1.

Analysis of GUS activity in transgenic Arabidopsis lines under cold stress

GUS reporter activity was assayed via histochemical staining using a GUS Staining Kit (Biotopped, Beijing, China). Various tissues were collected from the *TgCAMTA1 promoter::GUS* and *TgCAMTA3 promoter::GUS* transgenic plants after low-temperature treatment for 24 h. The samples were subsequently immersed in GUS staining solution and incubated for 12–24 h at 37 °C. When the samples were placed in 70% ethanol, the removal of chlorophyll from the samples resulted in a white color, serving as the negative control [43]. Morphological observations were performed and documented by capturing images with a camera (SONY, Japan).

Statistical analysis

Statistically significant differences ($P < 0.05$) were determined via one-way ANOVA with *Duncan's test* in IBM SPSS Statistics 26. All the data are expressed as the means ± standard deviations (SDs). Three biological replicates of each sample were collected.

Results

Identification and characterization of TgCAMTAs in teak

To understand the functional implications of the TgCAMTAs in teak, conserved domains in the protein sequences obtained through two methods were identified using NCBI-CDD. Six *CAMTA* genes were identified in the teak genome and named *TgCAMTA1*, *TgCAMTA2*, *TgCAMTA3*, *TgCAMTA4*, *TgCAMTA5*, and *TgCAMTA6* on the basis of their chromosomal positions. The amino acid numbers of the six CAMTA proteins ranged from 924 aa (TgCAMTA6) to 1098 aa (TgCAMTA3), with isoelectric points ranging from 5.49 (TgCAMTA5) to 7.55 (TgCAMTA1), and all the proteins had molecular weights exceeding 100 kDa (Table 1). The hydrophilicity index is a crucial indicator of protein hydrophobicity; the hydrophilicity index of TgCAMTAs ranged from 0.601 (TgCAMTA3) to -0.368 (TgCAMTA1), indicating that all the TgCAMTAs were hydrophilic. Further subcellular localization prediction demonstrated that all the TgCAMTAs were nucleus-localized proteins (Table 1).

Phylogenetic analysis of TgCAMTAs

To elucidate the functional and evolutionary relationships of CAMTA proteins in teak, multiple sequence alignment was conducted using 34 CAMTA protein sequences from diverse species. The phylogenetic tree was constructed via ML method. The 34 protein sequences were derived from teak (6 TgCAMTAs), Arabidopsis (6 AtCAMTAs), poplar (7 PtCAMTAs), and wheat (15 TaCAMTAs) (Fig. 1). On the basis of sequence similarity, these CAMTA proteins were classified into four subfamilies, with subfamily I comprising 7 proteins (2 TaCAMTA, 2 PtCAMTA, 2 AtCAMTA, and 1 TgCAMTA), subfamily II consisting of 10 proteins (6 TaCAMTA, 2 PtCAMTA, 1 AtCAMTA, and 1 TgCAMTA), subfamily III comprising 9 proteins (7 TaCAMTA, 1 PtCAMTA, and 1 TgCAMTA), and subfamily IV containing 8 proteins (2 PtCAMTA, 3 AtCAMTA, and 3 TgCAMTA) (Fig. 1). Members of the TgCAMTA family were distributed across all four subgroups. The phylogenetic tree revealed proteins closely evolutionarily related to the teak CAMTA family (TgCAMTA1 and PtCAMTA5; TgCAMTA3, TgCAMTA4 and AtCAMTA3; TgCAMTA6 and AtCAMTA4; TgCAMTA2 and TaCAMTa4-D, TaCAMTa5-D; TgCAMTA5 and AtCAMTA1, AtCAMTA2) (Fig. 1), indicating a high degree of conservation in the evolution of the CAMTA family across different species.

The gene structure and motif composition of the CAMTA family in teak are relatively conserved

Exon-intron formations of the *CAMTA* genes were examined to enhance our understanding of *CAMTA* family gene development. The positions of exons and introns are usually used to analyze the evolutionary relationships between different members of a gene family. Using the GSDS 2.0 online platform, analysis was conducted on six *TgCAMTAs* in teak. We found that all the *TgCAMTAs* exhibited multiple exonic regions, with *TgCAMTA1*, *TgCAMTA4*, and *TgCAMTA6* containing 12 exons while *TgCAMTA2*, *TgCAMTA3*, and *TgCAMTA5* contained 13 exons each (Fig. 2A, B).

The motif analysis of the TgCAMTA family in teak revealed that the different family members shared the same 10 motif types, indicating a high degree of conservation in their motif composition and gene structure (Fig. 2C, E). The observed high level of motif and structural conservation across the gene family members implies their functional importance and evolutionary relatedness within the teak species.

To predict the potential functions of TgCAMTA proteins, the conserved domains were identified via the NCBI-CDD online website. The Calmodulin Target Database (http://calcium.uhnres.utoronto.ca/ctdb/ctdb/) was

Table 1 Information of *TgCAMTA* genes and their encoded protein identified in the teak genome

Gene Name	Gene ID	Protein Size(aa)	Molecular Weight(kDa)	pI	GRAVY	Localization prediction
TgCAMTA1	Tg04g00190	947	105.5	7.55	-0.368	Nucleus
TgCAMTA2	Tg04g01320	927	104.9	6.89	-0.526	Nucleus
TgCAMTA3	Tg07g14230	1098	123.3	5.57	-0.601	Nucleus
TgCAMTA4	Tg07g14240	1067	119.4	6.04	-0.57	Nucleus
TgCAMTA5	Tg08g13920	1026	114.9	5.49	-0.484	Nucleus
TgCAMTA6	Tg11g14450	924	103.4	5.83	-0.478	Nucleus

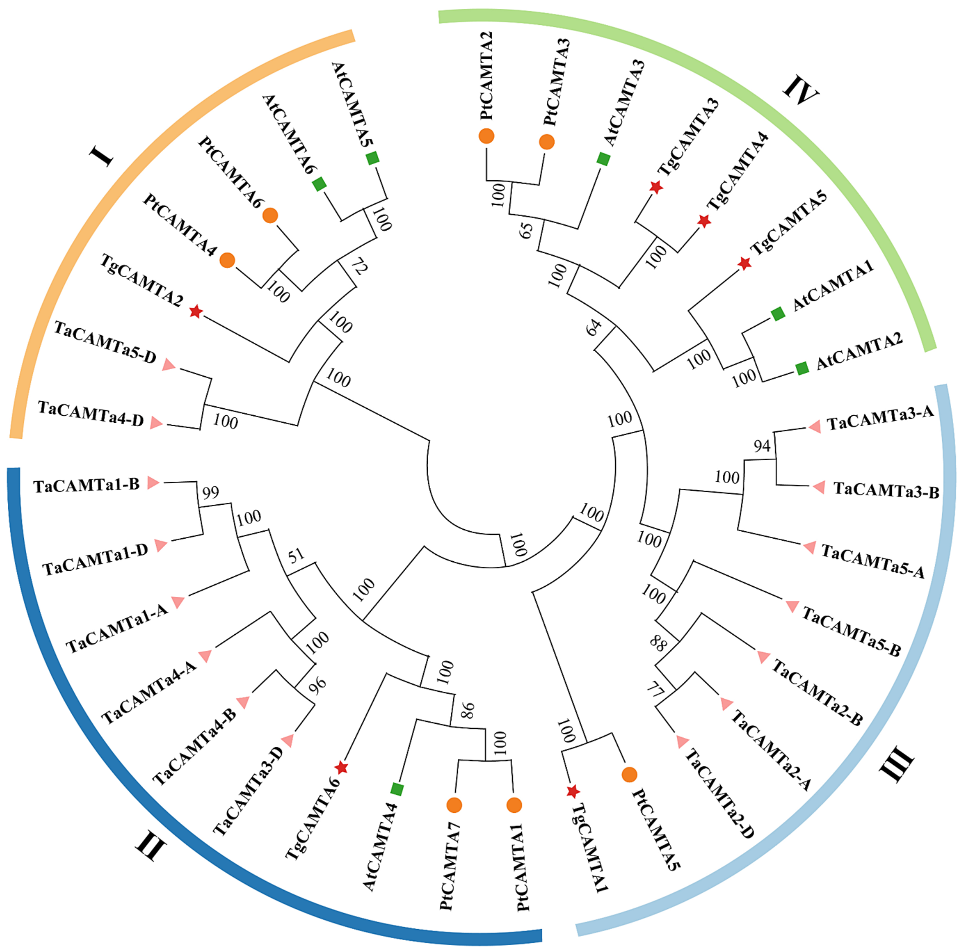

Fig. 1 Phylogenetic analysis of CAMTA proteins in different species was performed via MEGA11, and a phylogenetic tree of 34 CAMTA proteins was constructed, with the bootstrap value set to 1000. In the tree, the numerical values on the branches represent the branch support. The CAMTA proteins from teak are denoted by a pentagram, wheat CAMTA proteins by a triangle, Arabidopsis CAMTA proteins by a square, and poplar CAMTA proteins by a circle

used to predict the CaM-binding domains of the proteins [48]. The six TgCAMTA proteins exhibit a conserved domain architecture comprising the CG-1 domain, ANK repeat sequences, IQ motifs, and CaMBD (Fig. 2A, D). Furthermore, TgCAMTA2 and TgCAMTA4 in teak lack the conserved TIG domain (Fig. 2D), suggesting the need for further investigation to elucidate potential functional differences in these proteins.

Chromosomal location analysis of *TgCAMTAs*

The chromosomal distribution of the *TgCAMTA* gene family is depicted using relevant information from the teak genome database. As depicted in Fig. 3A, the six *CAMTA* genes in teak exhibited a nonuniform distribution on the pseudomolecules. Notably, Pseudomolecule_04 and Pseudomolecule_07 each harbor two members of the *TgCAMTA* gene family, representing the highest gene count on a pseudomolecule. Furthermore, *TgCAMTA1* and *TgCAMTA2* are located on Pseudomolecule_04, *TgCAMTA3* and *TgCAMTA4* are situated on Pseudomolecule_07, *TgCAMTA5* is positioned on Pseudomolecule_08, and *TgCAMTA6* is found on Pseudomolecule_11 (Fig. 3A).

Prediction of cis-acting elements in the *TgCAMTA* gene family

To gain an in-depth understanding of the specific regulation of stress response gene expression by the *TgCAMTA* gene family, the cis-acting elements in their promoter regions (the 2000 bp regions upstream from the transcription start sites) were identified and analyzed. The promoter regions of the *TgCAMTA* gene family in teak contain numerous cis-acting elements responsive to abiotic stress. Among these, apart from that of *TgCAMTA4*, the promoter regions of all the other genes include ABRE (Fig. 3B). All the *TgCAMTA* promoter regions possess the abiotic stress-responsive element MYC and ERE. The drought-responsive element MBS is found in the promoter regions of *TgCAMTA1*, *TgCAMTA2*, *TgCAMTA3* and *TgCAMTA5*. The gibberellin-responsive P-box

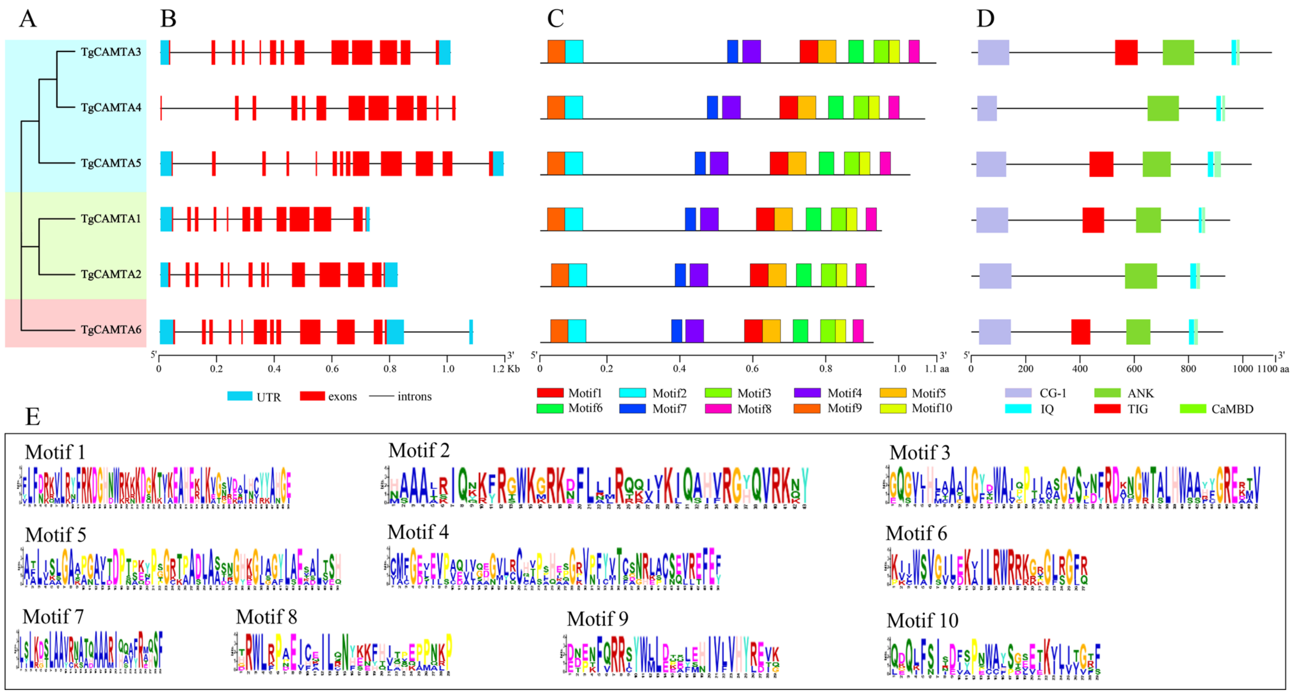

Fig. 2 Analysis of the gene structure, motifs, and domains of the teak CAMTA gene family. **A** Based on phylogenetic analysis, the 6 TgCAMTAs were divided into three subfamilies. **B** The exon–intron structure of the *TgCAMTA* genes was analyzed via the online tool GSDS. The lengths of exons and introns for each *TgCAMTA* gene are shown proportionally. The blue boxes, red boxes, and black lines represent untranslated regions (UTRs), exons, and introns, respectively. **C** Distribution of conserved motifs in TgCAMTA proteins. Different conserved motifs are shown in different colored boxes. **D** Different colored boxes represent different structural domains of TgCAMTA proteins, and the black lines represent amino acid sequences. **E** Sequence logos of conserved motifs. The overall height at each position represents the conservation level, and the height of each letter is proportional to its frequency of occurrence

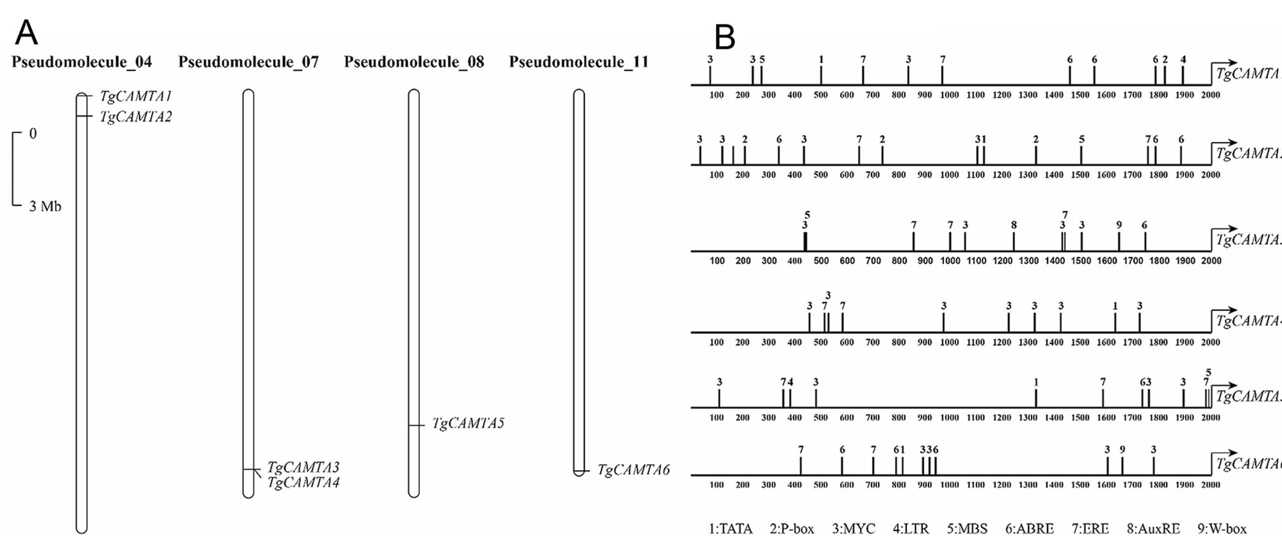

Fig. 3 **A** Distribution of *TgCAMTAs* on the chromosomes of teak. Chromosome localization of *TgCAMTAs* in teak, where chromosomes are named Pseudomolecule_01 to Pseudomolecule_19 according to the assembly of the teak genome. The scale is in megabases (Mb). **B** Cis-acting elements of the *TgCAMTA* genes. TATA, RNA polymerase binding site; P-box, gibberellin response element; MYC, abiotic stress response element; LTR, low-temperature response element; MBS, drought response element; ABRE, ABA response element; ERE, ethylene response element; AuxRE, auxin response element; W-box, jasmonate response element

element is present in the promoter regions of *TgCAMTA1* and *TgCAMTA2*. The *TgCAMTA3* and *TgCAMTA6* promoter regions contain the auxin-responsive W-box element (Fig. 3B). However, another auxin-responsive element, AuxRE, is found only in the *TgCAMTA3* promoter region. Moreover, the low-temperature-responsive (LTR) element is present in the promoter regions of *TgCAMTA1* and *TgCAMTA5* (Fig. 3B). These findings suggest the significant roles of the *TgCAMTA* family in plant growth, stress response, and hormone signaling pathways.

Prediction of the 3D structure of TgCAMTAs proteins

Proteins require specific higher-order structures to carry out their functions. The 3D structures of the TgCAMTA proteins were predicted via the SWISS-MODEL online tool and visualized via PyMOL v2. All the TgCAMTA proteins contain the CG-1 DNA-binding domain, which is located independent of the peptide chain N-terminus, facilitating better transcription factor binding (Fig. 4). The Ca^{2+}-dependent CaMBD domain is adjacent to the IQ motif and exposed on the outer surface. The ANK repeats are associated with protein interactions in the TgCAMTA family. ANK repeat sequences fold to form 6–7 compact helical bundles (TgCAMTA1, TgCAMTA5, and TgCAMTA6 contain 6 helical bundles, whereas TgCAMTA2, TgCAMTA3, and TgCAMTA4 contain 7 helical bundles). TgCAMTA1, TgCAMTA3, TgCAMTA5, and TgCAMTA6 all possess the TIG domain (Fig. 4), indicating potential functional diversity.

Analysis of the expression patterns of *TgCAMTAs* in teak

To investigate the differential tissue expression patterns of *TgCAMTAs*, transcriptomic data of the teak *CAMTA* gene family in various tissues were analyzed (Supplementary Fig. 2). These results indicate that *TgCAMTA* genes are expressed across different tissue types. Among them, *TgCAMTA3* and *TgCAMTA6* presented the highest expression in leaves. Additionally, all the *TgCAMTA* family genes presented the lowest expression in seeds (Supplementary Fig. 2). This evidence suggests that the

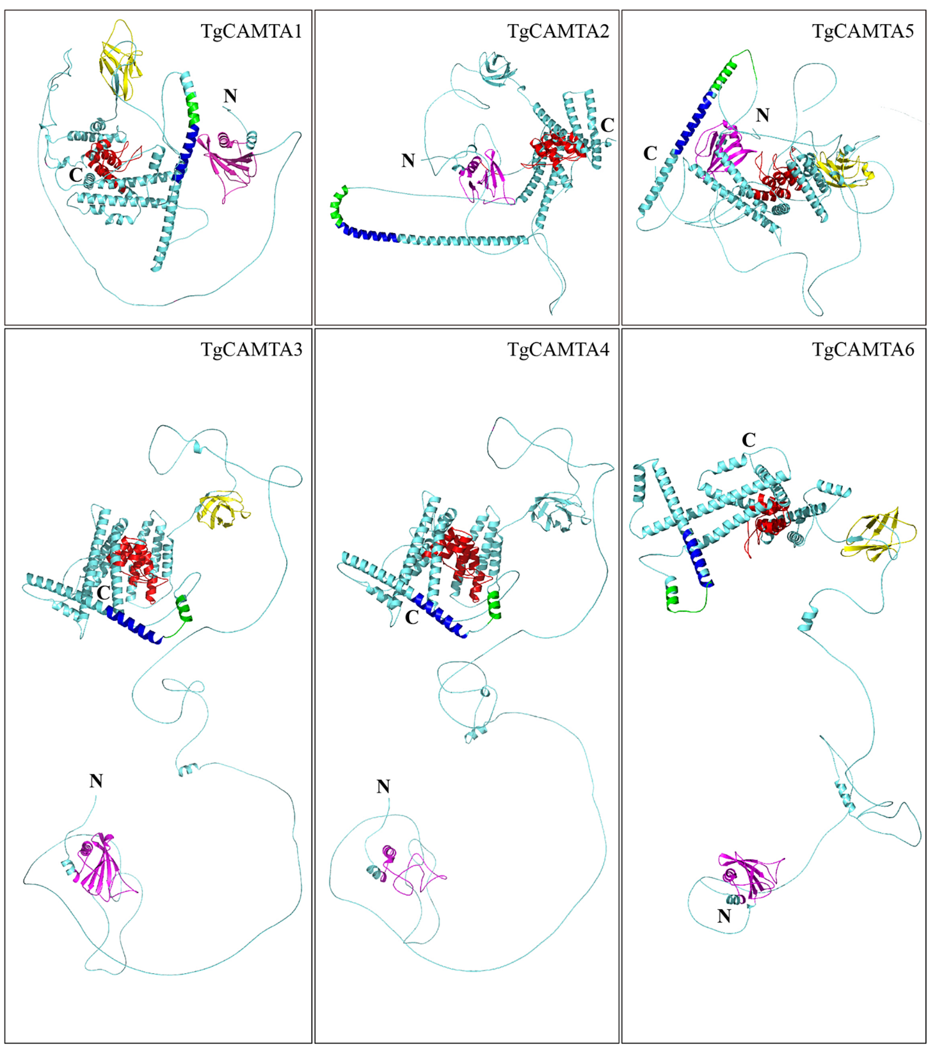

Fig. 4 3D structure model of the TgCAMTA protein. Structure models were predicted with SWISS-MODEL and visualized with PyMOL software. The different structural domains of the TgCAMTA proteins are marked with different colors. Yellow, TIG domain; red, ANK domain; pink, CG-1 domain; green, CaMBD domain; blue, IQ domain; N, N-terminus of the peptide chain; C, C-terminus of the peptide chain

TgCAMTA family plays crucial roles in plant growth and development processes.

Furthermore, to further explore the potential roles of the *TgCAMTA* gene family in the response to low-temperature stress, seedlings were subjected to low-temperature stress at 4 °C. The relative expression levels of teak *TgCAMTA* genes in the roots, stems, and leaves at different time points under low-temperature stress were analyzed via RT−qPCR. Under normal conditions without low-temperature stress, *TgCAMTA1* presented the highest expression in stems, followed by leaves, and the lowest expression in roots (Fig. 5). The highest expression of *TgCAMTA2* was detected in the roots, whereas relatively low expression was detected in the stems and leaves. The *TgCAMTA3*, *TgCAMTA4*, *TgCAMTA5*, and *TgCAMTA6* genes all presented the highest expression in leaves and the lowest expression in roots. Following low-temperature stress treatment, the expression of teak *TgCAMTA* genes in the roots, stems, and leaves varied over time. Specifically, the expression of *TgCAMTA1* shifted from being highest in stems to being highest in leaves, except after 24 h of treatment (Fig. 5). The expression of *TgCAMTA2* changed from being highest in roots to being highest in stems after 24 h of treatment. The highest expression of *TgCAMTA5* was detected in stems at 1 h, 3 h, and 24 h. The highest expression of *TgCAMTA4* and *TgCAMTA6* was detected in the roots at 3 h and 1 h, respectively (Fig. 5). These results indicate significant variations in the expression of teak *TgCAMTA* family genes under low-temperature stress.

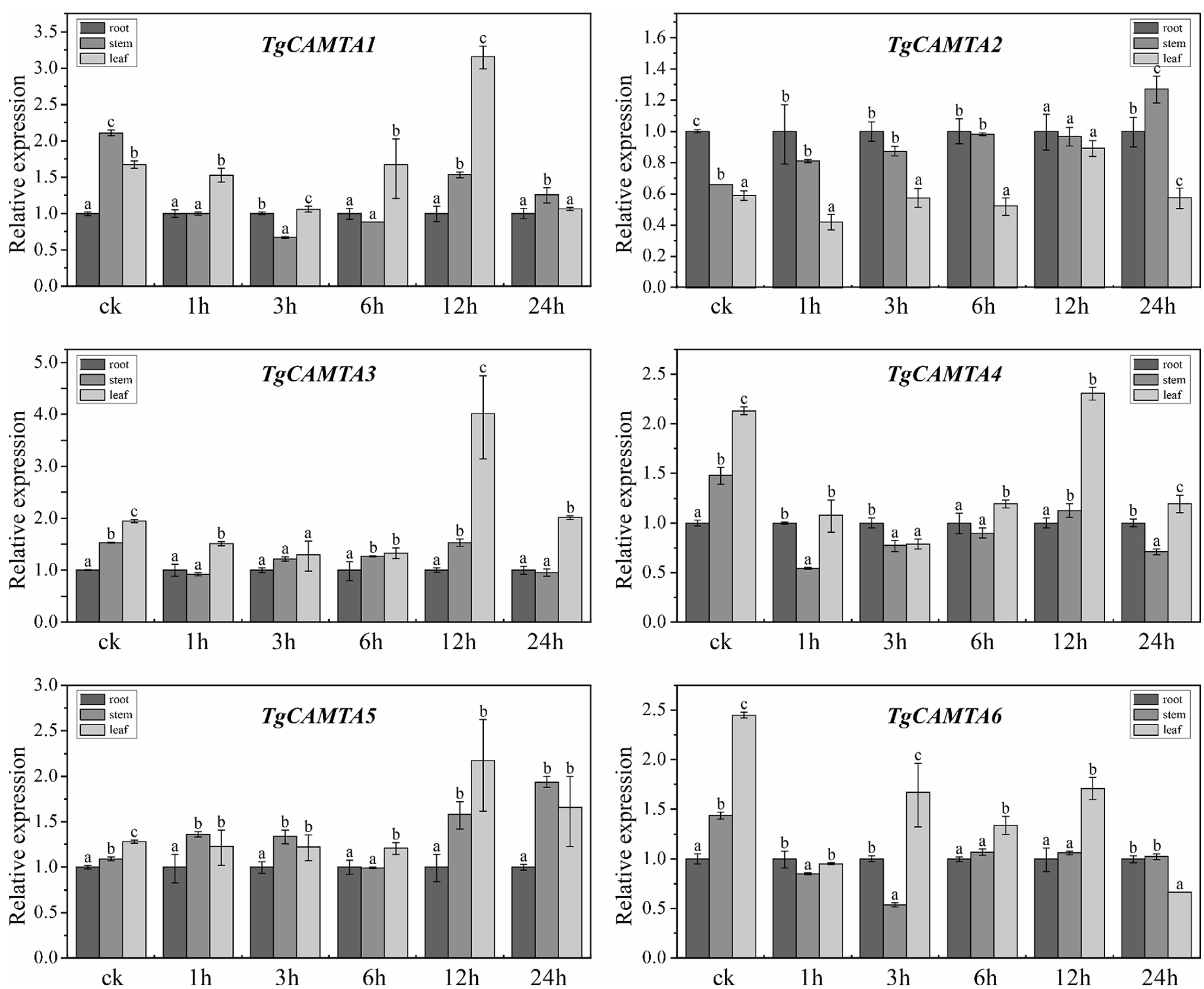

Fig. 5 Changes in the relative expression of *TgCAMTA* genes in the roots, stems, and leaves of teak at different time points of low-temperature stress. The data represent the mean values of three biological replicates with standard deviations (SDs). "ck" represents teak seedlings without low-temperature stress treatment. The letters "a", "b", and "c" indicate significant differences in *TgCAMTA* gene expression among the roots, stems, and leaves of teak ($n=9$, *Duncan's t test*)

Promoter regions of *TgCAMTA1* and *TgCAMTA3* contain cold-responsive elements

In teak, the *CAMTA* gene family exhibits considerable variation under conditions of low-temperature stress (Fig. 5). Cis-acting element analysis revealed the presence of the LTR element exclusively in the promoter regions of *TgCAMTA1* and *TgCAMTA5* (Fig. 3B). To further determine the functionality of the LTR element in the response to low-temperature stress, transgenic Arabidopsis plants transformed with a vector harboring the *TgCAMTA1 promoter::GUS* gene or the *TgCAMTA3 promoter::GUS* gene were produced. A histochemical staining assay revealed that before 4 °C treatment, minimal blue staining was observed in the leaves of the transgenic plants harboring the *TgCAMTA1 promoter::GUS* gene, with no detectable blue staining in the remaining plant tissues (Fig. 6A). However, after 24 h of 4 °C treatment, GUS activity was detected in the *TgCAMTA1 promoter::GUS* and *TgCAMTA3 promoter::GUS* transgenic Arabidopsis plants, and all of the plants uniformly transformed into a blue hue (Fig. 6A). Furthermore, the relative expression levels of the GUS gene were examined during the 24-h low-temperature stress treatment. Without treatment, significant changes in GUS gene expression were not detected in the transgenic Arabidopsis. However, the *TgCAMTA1 promoter::GUS* transgenic plants presented the highest expression at 6 h, whereas the *TgCAMTA3 promoter::GUS* transgenic plants presented the highest expression at 3 h (Fig. 6B). These data indicate that the LTR cis-acting element is not essential for the low-temperature stress response of TgCAMTA3.

Subcellular localization of TgCAMTA1 and TgCAMTA3 in tobacco leaves

Subcellular localization analysis via the Plant-mPLoc website revealed that both TgCAMTA1 and TgCAMTA3 are localized within the cell nucleus (Table 1). To verify this prediction, the subcellular location was checked in tobacco leaves by overexpression of *TgCAMTA1* and *TgCAMTA3* (nonterminator) via the pbi121 GFP vector driven by the CaMV35S promoter. Moreover, we employed the previously reported nuclear localization signal (NLS) protein SV40 as a marker. The red fluorescence signal of the nuclear marker colocalized with the green fluorescence signal in cells transformed with the TgCAMTA1 and TgCAMTA3 fusion protein constructs at the nuclear position. In contrast, the cells transformed with the control vector presented fluorescence in both the cell membrane and the nucleus, suggesting that TgCAMTA1 and TgCAMTA3 are localized in the nucleus (Fig. 7).

Overexpression of *TgCAMTA1* and *TgCAMTA3* in *Arabidopsis thaliana* confers tolerance to cold stress

To investigate the response of *TgCAMTA1* and *TgCAMTA3* to low-temperature stress, we selected two pairs of stable overexpression lines, OE1-5/OE1-8 and OE3-1/OE3-9 in Arabidopsis, on the basis of agarose gel electrophoresis and RT−qPCR results for further analysis (Supplementary Fig. 3A, B, Fig. 8A).

To confirm the functionality of *TgCAMTA1* and *TgCAMTA3* under low-temperature conditions, WT and transgenic Arabidopsis lines were grown vertically on 1/2 MS medium for 6 days, followed by cold treatment. Before low temperature stress treatment, no significant

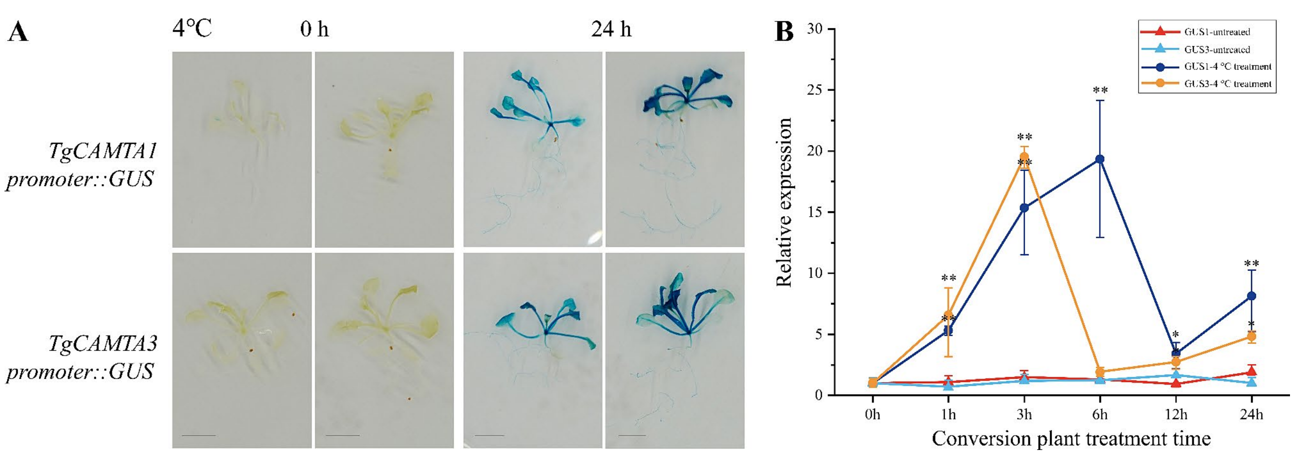

Fig. 6 GUS expression pattern analysis. **A** Histochemical staining for determining GUS activity in plants harboring the *TgCAMTA1 promoter::GUS* gene and the *TgCAMTA3 promoter::GUS* gene after treatment at 4 °C for 0 h or 24 h. **B** RT–qPCR analysis of GUS expression after cold treatment for 0 h, 1 h, 3 h, 6 h, 12 h, or 24 h. We administered low-temperature stress treatment to two-week-old seedlings grown on MS medium. For the cold treatment of seedlings under normal medium culture conditions at 4 °C for 24 h, we employed seedlings grown on MS medium as a negative control under normal temperature conditions. The data are shown as a line chart, with the whiskers representing the means ± SDs. Statistically significant differences were determined by one-way ANOVA with *Duncan's test*. '*' and '**' indicate p values < 0.05 and < 0.01, respectively. GUS1, *TgCAMTA1 promoter::GUS*; GUS3, *TgCAMTA3 promoter::GUS*

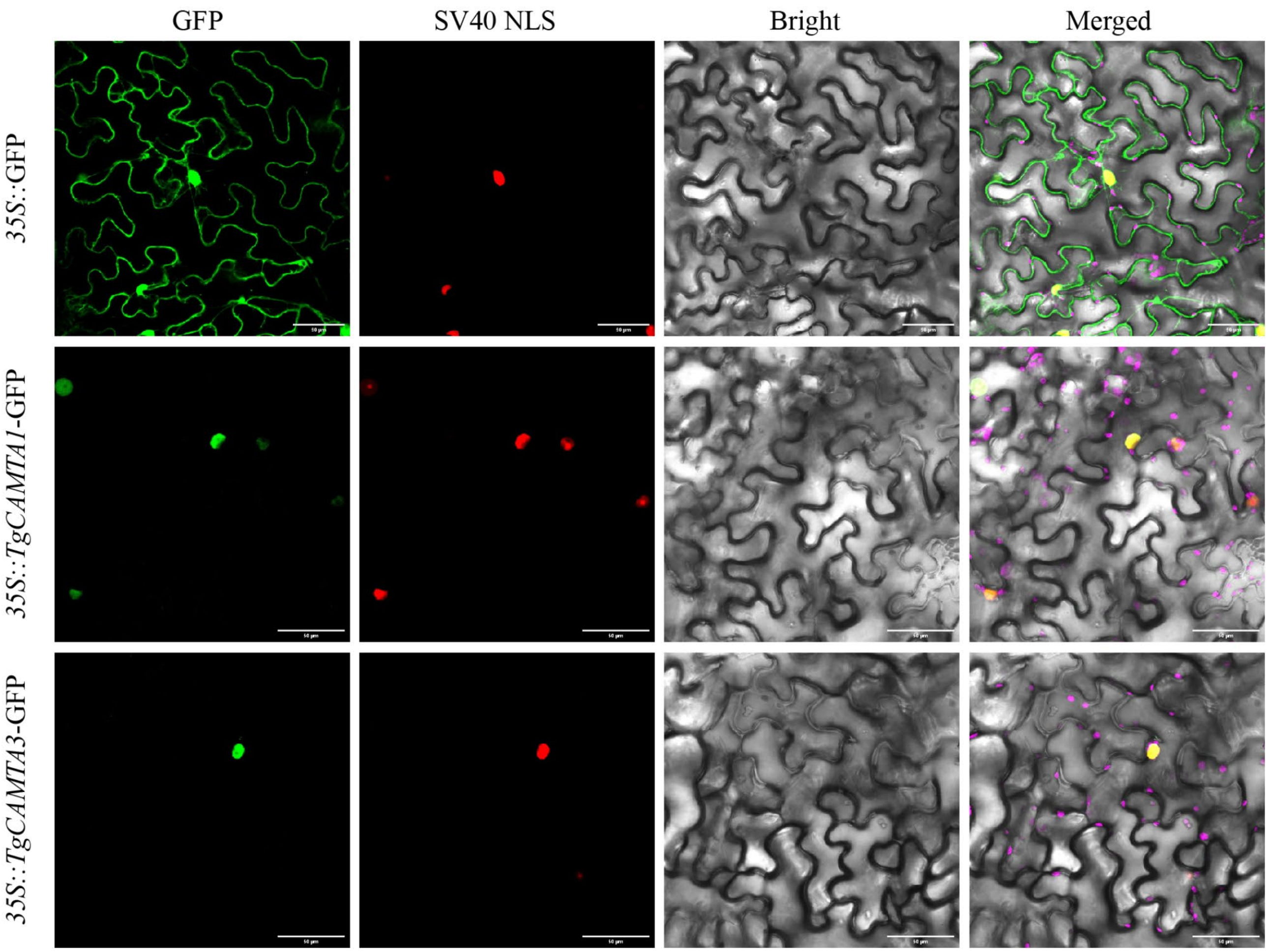

Fig. 7 Subcellular location of TgCAMTA1 and TgCAMTA3 in tobacco. Green represents the GPF fluorescence emitted after excitation at a wavelength of 488 nm, red denotes the signal of the nuclear localization marker emitted after excitation at 561 nm, and pink symbolizes the spontaneous chloroplast fluorescence emitted after excitation at 640 nm. Bar = 50 μm

differences were detected between the WT and transgenic lines (Fig. 8B). After low temperature stress treatment, the WT plants presented slow growth, whereas the two Arabidopsis lines overexpressing of *TgCAMTA1* and *TgCAMTA3* presented greater leaf numbers than the WT plants did. We subsequently conducted a statistical analysis of the primary root lengths of the transgenic lines and WT plants, revealing that after low-temperature treatment, the primary root lengths of the four overexpression lines were significantly greater than those of the WT plants (Fig. 8C). Overall, the results revealed that the transgenic plants were more tolerant than the WT plants to low temperature, suggesting that the *TgCAMTA1* and *TgCAMTA3* genes might play a key role in the plant response to cold stress tolerance.

In Arabidopsis, members of the *CAMTA* gene family can respond to low-temperature stress through the CBF cold signaling pathway [14]. We examined the expression of key genes in the CBF cold signaling pathway, including *AtCBF1*, *AtCBF2*, and *AtCBF3*. The expression levels of these genes were assessed via RT−qPCR in the transgenic lines and WT seedlings. In the *TgCAMTA1* overexpression line, before low temperature stress treatment, the expression levels of *AtCBF1*, *AtCBF2*, and *AtCBF3* in lines OE1-5 were significantly greater than those in the WT and OE1-8 lines (Fig. 8D). After low temperature stress treatment, the expression levels of *AtCBF1*, *AtCBF2*, and *AtCBF3* in lines OE1-5 and OE1-8 were lower than those in the WT. Conversely, in the *TgCAMTA3* overexpression line, before low temperature stress treatment, the expression levels of *AtCBF* in lines OE3-1 and OE3-9 were 1–3 times greater (*AtCBF1* and *AtCBF2*) than those in the WT or no significant difference (*AtCBF3*) (Fig. 8D). After low temperature stress treatment, the expression levels of *AtCBF1*, *AtCBF2* and *AtCBF3* in lines OE3-1 and OE3-9 were 5–10 times greater than those in the WT (Fig. 8D). Furthermore, the expression levels downstream of the CBF cold stress pathway of *AtCOR15A* and *AtCOR47* significantly increased in all four overexpression lines under low-temperature stress (Fig. 8D). These results

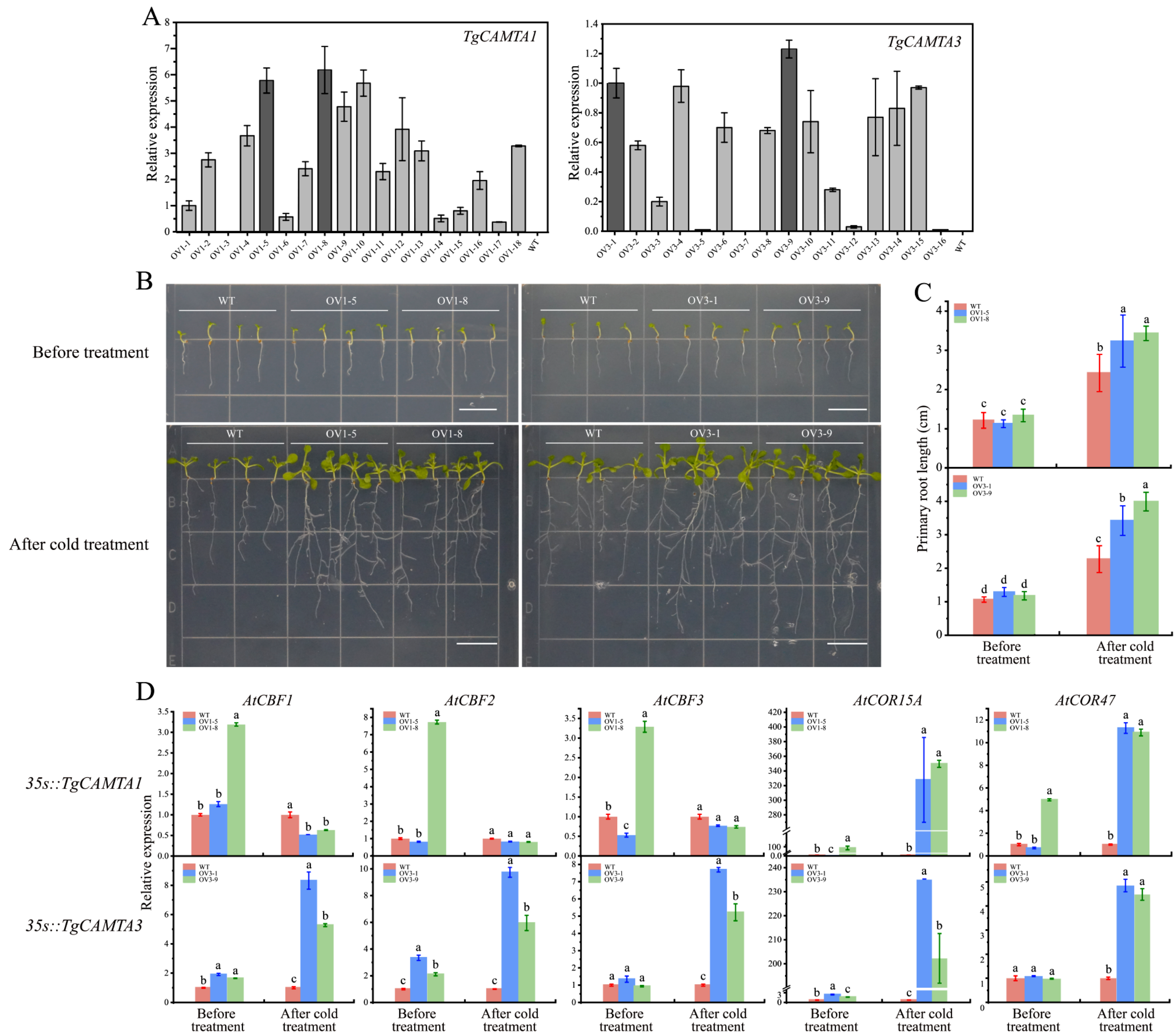

Fig. 8 Growth status and gene expression in transgenic lines of *TgCAMTA1* and *TgCAMTA3* were evaluated under conditions of cold stress. **A** The expression of *TgCAMTA1* and *TgCAMTA3* was examined in T2 transgenic lines and wild-type Arabidopsis seedlings. **B**, **C** The growth status and primary root length of the transgenic lines and wild type were measured on 1/2 MS medium before or after low-temperature conditions with vertical growth. **D** Expression levels of CBF cold signaling pathway genes (*AtCBF1/2/3*), *AtCOR15A*, and *AtCOR47* in the transgenic lines and wild-type Arabidopsis. The data are shown as histograms, with whiskers representing the means ± SDs. Statistically significant differences ($P < 0.05$) were determined via one-way ANOVA with *Duncan's test* and are indicated with different letters (a-d). Bar = 1 cm

indicate that *TgCAMTA3* interacts with the endogenous CBF pathway in *Arabidopsis* under low-temperature conditions.

Discussion

Since their initial discovery, the structure and function of CAMTA transcription factors have been extensively studied in various species [6, 11, 15]. However, research on CAMTAs in woody plants, particularly in the Labiatae family, is still in its early stages. In this study, six *CAMTA* gene family members were identified from the teak genome (Table 1), which is consistent with the number of CAMTA family members in Arabidopsis [10] but fewer than the 15 members in upland cotton, 7 members in poplar, and 15 members in both wheat and soybean [8, 49–52]. We speculate that the main reason for this difference is that teak and *Arabidopsis* are diploid, poplar is triploid, upland cotton and soybean are tetraploid, and wheat is hexaploid. Subcellular localization information is crucial for understanding the potential functions of proteins in various biological processes [53]. Previous studies have indicated that CAMTA proteins function primarily as calmodulin-binding transcription factors, regulating the expression of other genes [20, 49].

Consistent with this, subcellular localization studies in tobacco revealed that the TgCAMTA1 and TgCAMTA3 proteins were both localized in the nucleus (Fig. 7).

The CaMBD structural domain within CAMTA proteins plays a crucial role in the transduction of Ca^{2+} signals [49]. The identified TgCAMTA protein members contain the conserved CaMBD structural domain, which functions primarily through Ca^{2+}-dependent binding with CaM. Additionally, the IQ structural domain represents a Ca^{2+}-independent mode of interaction with CaM [10, 54]. The highly conserved nature of the CaMBD and IQ domains in the teak CAMTA family members indicates that both modes of functionality may coexist. Moreover, all teak gene families have domain CG-1 domain, ANK repeat sequences, IQ motifs, and CaMBD (Fig. 2A, D), which is consistent with the typical domain composition of the CAMTA family reported in previous studies [8, 19]. Furthermore, the predicted three-dimensional structure of the TgCAMTA proteins demonstrated that the CaMBD and IQ domains are adjacent and exposed on the outer surface of the protein tertiary structure, further supporting the notion that CAMTA proteins function in two distinct ways. However, different CAMTA proteins, such as *HbCAMTA3* from rubber, lack the IQ domain, thus relying solely on Ca^{2+}-dependent mechanisms for their functionality [50]. Therefore, further detailed investigations are needed to confirm the precise nature of the CAMTA functionality in teak.

Gene expression patterns provide essential clues for understanding biological functions [55, 56]. In poplar, the *CAMTA* gene family is expressed in roots, stems, and leaves. Compared with other genes, *HbCAMTA1* is expressed at higher levels in latex, leaves, roots, bark, pistillate flowers, and staminate flowers [50]. Transcriptomic data indicate that *TgCAMTAs* are expressed in various tissues of teak, with most *TgCAMTA* genes highly expressed in leaves and stems (Supplementary Fig. 2), suggesting their significant roles in the physiological and biochemical processes occurring in teak stems and leaves. Furthermore, following cold stress, the expression of the *TgCAMTA* family genes in teak is either upregulated or downregulated in roots, stems, and leaves (Fig. 5), indicating their crucial functions in different tissues in response to low-temperature stress.

Cis-acting elements play crucial roles in plant responses to low-temperature stress [39, 40, 42]. In *Dimocarpus longan*, the expression levels of *DlCBF1/2* significantly increased after cold stress, peaking at 9 h. Promoter analysis of *DlCBF1/2* revealed the presence of multiple LTR cis-elements (CCGAAA) [15]. In addition, the MYC recognition site (CANNTG) acts as a binding site for cold-induced (ICE)-like proteins [57]. Both the *TgCAMTA1 promoter::GUS* gene construct containing the LTR and the *TgCAMTA3 promoter::GUS* gene construct lacking the LTR in transgenic *Arabidopsis* presented significant increased GUS activity after cold stress (Fig. 6). It is speculated that the MYC region in the promoters of *TgCAMTA1* and *TgCAMTA3* plays a critical role in the response to low-temperature stress in transgenic *Arabidopsis*. These findings are consistent with the function of the MYC cis-acting elements in the *MeCBF1* promoter region in cassava [42]. Additionally, GUS gene expression was detected in the transgenic *Arabidopsis* lines without cold treatment, with minimal blue coloration observed in the leaves of the *TgCAMTA1 promoter::GUS* line, suggesting the presence of other transcription factors that bind to its cis-acting elements under normal conditions (Fig. 6).

In previous studies, *CAMTA* genes were shown to play crucial roles in the response to low-temperature stress [13, 18, 58]. Compared with WT plants, the Arabidopsis *camta3/sr1* mutant was highly susceptible to temperature reduction and exhibited growth inhibition when the temperature was decreased to 20 °C [59]. Additionally, the DREB/CBF pathway is a well-recognized key regulatory pathway for low-temperature signaling in plants [60], with a CAMTA protein acting as a positive regulator that controls the expression of *CBF1* and *CBF2* [14, 61]. In our study, compared with WT Arabidopsis, transgenic Arabidopsis overexpressing *TgCAMTA1* and *TgCAMTA3* presented increased leaf numbers and primary root lengths after low-temperature stress (Fig. 8B, C). The expression levels of *AtCBF1/2/3*, *AtCOR15A*, and *AtCOR47* significantly increased in the *TgCAMTA3*-overexpressing plants after low-temperature stress (Fig. 8E). In agreement, overexpression of longan *DlCBF1/2/3* in Arabidopsis also enhances the expression of cold-responsive genes *AtRD29A*, *AtCOR15A*, and *AtCOR47*. We propose that *TgCAMTA3* in Arabidopsis responds to low-temperature stress through the CBF pathway. Furthermore, the expression levels of *AtCOR15A* and *AtCOR47* significantly increased in *TgCAMTA1*-overexpressing plants after low-temperature stress, indicating that the response of *TgCAMTA1* to low-temperature stress can regulate the expression *AtCOR15A* and *AtCOR47* through alternative pathways. *AtCOR15* and *AtCOR47*, as members of the LEA family, have been reported to be highly expressed in response to low-temperature stress in plants, thereby enhancing their freezing tolerance [62, 63]. In our study, we found that *TgCAMTA1/3* can increase the expression levels of *AtCOR15A* and *AtCOR47* through various pathways; however, the regulatory network associated with these genes remains unclear and requires further validation.

Conclusions

In conclusion, our study identified and analyzed 6 *TgCAMTA* genes in teak. They were classified into four groups, all of which contained the CaMBD domain. Expression pattern analyses indicated that all *TgCAMTAs* were responsive to cold stress. Examination of GUS activity revealed the presence of cold-responsive elements in the promoter regions of *TgCAMTA1* and *TgCAMTA3*. In addition, overexpression Arabidopsis of *TgCAMTA1* and *TgCAMTA3* increased tolerance to low temperature. These findings suggest that *TgCAMTAs* genes could play a critical role in response to cold stress, and have a great potential to be used as the candidate genes for the genetic engineering of teak with improved tolerance to low-temperature stress.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12870-024-05788-w.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

Acknowledgements
The authors appreciate those contributors who make related genome and transcriptome data accessible in public databases. Our gratitude extends to Liwen Shan for her invaluable recommendations concerning the colour coordination of the illustrations in the manuscript.

Author contributions
G.H. and T.F. conceived the original screening and research plans; W.Z., J.D., R.J., X.W., T.F., and G.H. performed the experiments and analyzed the data; W.Z. and G.H. wrote the article with contributions of all the authors; G.H. agrees to serve as the author responsible for contact and ensures communication.

Funding
This work was supported by the National Key Research and Development Program of China (2024YFD2200102); Guangdong Forestry Administration (Collection, preservation, propagation and utilization of cold-tolerant germplasm resources on teak).

Data availability
All data supporting the findings of this study are available within the paper and within its supplementary data published online.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]Research Institute of Tropical Forestry, Chinese Academy of Forestry, Guangzhou 520521, China
[2]The Forestry Development Service Center of Ganzhou City, Ganzhou 341000, China
[3]Guangdong Forest Resources Conservation Center, Guangzhou 520521, China

Received: 25 September 2024 / Accepted: 6 November 2024
Published online: 10 January 2025

References

1. Clapham DE. Calcium signaling. Cell. 1995;80:259–68.
2. Tuteja N, Mahajan S. Calcium Signaling Network in plants: an overview. Plant Signal Behav. 2007;2:79–85.
3. Hashimoto K, Kudla J. Calcium decoding mechanisms in plants. Biochimie. 2011;93:2054–9.
4. Zeng HQ, Wang GP, Wang HZ et al. Functions of calmodulin-binding transcription activators (CAMTAs) in plants. Plant Physiol. 2015;51 (5): 633–41.
5. Dodd AN, Kudla J, Sanders D. The Language of Calcium Signaling. Annu Rev Plant Biol. 2010;61:593–620.
6. Lorenzo O. bZIP edgetic mutations: at the frontier of plant metabolism, development and stress trade-off. J Exp Bot. 2019;70:5517–20.
7. Ye Y, Wu K, Chen J, Liu Q, Wu Y, Liu B, et al. OsSND2, a NAC family transcription factor, is involved in secondary cell wall biosynthesis through regulating MYBs expression in rice. Rice. 2018;11:36.
8. Identification. and functional analysis of citrus CAMTA genes. Dissertation, Southwest University.2019.
9. Rahman H, Yang J, Xu Y-P, Munyampundu J-P, Cai X-Z. Phylogeny of Plant CAMTAs and Role of AtCAMTAs in Nonhost Resistance to *Xanthomonas oryzae pv. Oryzae*. Front Plant Sci. 2016;7.
10. Bouché N, Scharlat A, Snedden W, Bouchez D, Fromm H. A Novel Family of Calmodulin-binding transcription activators in multicellular organisms. J Biol Chem. 2002;277:21851–61.
11. Finkler A, Ashery-Padan R, Fromm H, CAMTAs. Calmodulin-binding transcription activators from plants to human. FEBS Lett. 2007;581:3893–8.
12. Liu M, Yang G, Zhou W, Wang X, Han Q, Wang J, et al. Transcriptomic analysis reveals CBF-Dependent and CBF-Independent pathways under low-temperature stress in Teak (*Tectona grandis*). Genes. 2023;14:2098.
13. Kim Y, Park S, Gilmour SJ, Thomashow MF. Roles of CAMTA transcription factors and salicylic acid in configuring the low-temperature transcriptome and freezing tolerance of Arabidopsis. Plant J. 2013;75:364–76.
14. Kidokoro S, Yoneda K, Takasaki H, Takahashi F, Shinozaki K, Yamaguchi-Shinozaki K. Different Cold-Signaling pathways function in the responses to Rapid and Gradual decreases in temperature. Plant Cell. 2017;29:760–74.
15. Yang X, Wang R, Jing H, Chen Q, Bao X, Zhao J, et al. Three novel C-Repeat binding factor genes of Dimocarpus longan regulate cold stress response in Arabidopsis. Front Plant Sci. 2020;11:1026.
16. Hernández-Sánchez I, Rindfleisch T, Alpers J, Dulle M, Garvey CJ, Knox-Brown P, et al. Functional in vitro diversity of an intrinsically disordered plant protein during freeze–thawing is encoded by its structural plasticity. Protein Sci. 2024;33:e4989.
17. Wang X, Li Z, Shi Y, Liu Z, Zhang X, Gong Z, et al. Strigolactones promote plant freezing tolerance by releasing the WRKY41 -mediated inhibition of *CBF / DREB1* expression. EMBO J. 2023;42:e112999.
18. Pandey N, Ranjan A, Pant P, Tripathi RK, Ateek F, Pandey HP, et al. *CAMTA 1* regulates drought responses in *Arabidopsis thaliana*. BMC Genomics. 2013;14:216.
19. Yang F. Genome-wide identification and expression analysis of the calmodulin-binding transcription activator (CAMTA) gene family in wheat (*Triticum aestivum* L). BMC Genet. 2020;21:105.
20. Wei M, Xu X, Li C. Identification and expression of CAMTA genes in *Populus trichocarpa* under biotic and abiotic stress. Sci Rep. 2017;7:17910.
21. Vongkhamho S, Imaya A, Takenaka C, Yamamoto K, Yamamoto H. Correlations among Tree Quality, stand characteristics, and site characteristics in Plantation Teak in mountainous areas of Lao PDR. Forests. 2020;11:777.
22. Pérez Cordero L, Kanninen M. Heartwood, sapwood and bark content, and wood dry density of young and mature teak (*Tectona grandis*) trees grown in Costa Rica. Silva Fenn. 2003;37.
23. Ma HM, Liang KN, Zhou ZZ. Research and development of teak in China. For Res. 2003;16(6):768–73.
24. Zhao D, Hamilton JP, Bhat WW, Johnson SR, Godden GT, Kinser TJ et al. A chromosomal-scale genome assembly of Tectona grandis reveals the

importance of tandem gene duplication and enables discovery of genes in natural product biosynthetic pathways. GigaScience. 2019;8.
25. Lamesch P, Berardini TZ, Li D, Swarbreck D, Wilks C, Sasidharan R, et al. The Arabidopsis Information Resource (TAIR): improved gene annotation and new tools. Nucleic Acids Res. 2012;40:1202–10. Database issue:D.
26. Paysan-Lafosse T, Blum M, Chuguransky S, Grego T, Pinto BL, Salazar GA, et al. InterPro in 2022. Nucleic Acids Res. 2023;51:D418–27.
27. Potter SC, Luciani A, Eddy SR, Park Y, Lopez R, Finn RD. HMMER web server: 2018 update. Nucleic Acids Res. 2018;46 Web Server issue:W200–4.
28. Qi D, Wang L, Liang M, Zhang Q, Tang X, Geng B, et al. Genome-wide analyses of metal tolerance protein genes in apple (*Malus domestica*): identification, characterization, expression and response to various metal ion stresses. Environ Exp Bot. 2022;201:104948.
29. Yang M, Derbyshire MK, Yamashita RA, Marchler-Bauer A. NCBI's conserved domain database and Tools for Protein Domain Analysis. Curr Protocols Bioinf. 2020;69:e90.
30. Wilkins MR, Gasteiger E, Bairoch A, Sanchez J-C, Williams KL, Appel RD, et al. Protein Identification and Analysis Tools in the ExPASy server. 2-D Proteome Analysis Protocols. New Jersey: Humana; 1998. pp. 531–52.
31. Chou K-C, Shen H-B. Plant-mPLoc: A Top-Down Strategy to augment the Power for Predicting Plant Protein Subcellular Localization. PLoS ONE. 2010;5:e11335.
32. Goodstein DM, Shu S, Howson R, Neupane R, Hayes RD, Fazo J et al. Phytozome: a comparative platform for green plant genomics. Nucleic Acids Res. 2014;40:D1178–86.
33. Yates AD, Allen J, Amode RM, Azov AG, Barba M. Ensembl genomes: an expanding genome resource for non-vertebrates. Nucleic Acids Res. 2022;50:D996–1003.
34. Tamura K, Stecher G, Kumar S. MEGA11: Molecular Evolutionary Genetics Analysis Version 11. Mol Biol Evol. 2021;38:3022–7.
35. Hu B, Jin J, Guo A-Y, Zhang H, Luo J, Gao G. GSDS 2.0: an upgraded gene feature visualization server. Bioinformatics. 2015;31:1296–7.
36. Waterhouse A, Bertoni M, Bienert S, Studer G, Tauriello G, Gumienny R, et al. SWISS-MODEL: homology modelling of protein structures and complexes. Nucleic Acids Res. 2018;46:W296–303.
37. DeLano. WL, others. Pymol: an open-source molecular graphics tool. CCP4 Newsl Protein Crystallogr. 2002;40:82–92.
38. Chen C, Chen H, Zhang Y, Thomas HR, Frank MH, He Y, et al. TBtools: an integrative Toolkit developed for interactive analyses of big Biological Data. Mol Plant. 2020;13:1194–202.
39. Rong C, Zhang R, Liu Y, Chang Z, Liu Z, Ding Y, et al. Purine permease (PUP) family gene *PUP11* positively regulates the rice seed setting rate by influencing seed development. Plant Cell Rep. 2024;43:112.
40. Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, et al. PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in silico analysis of promoter sequences. Nucleic Acids Res. 2002;30:325–7.
41. Galeano E, Vasconcelos TS, Vidal M, et al. Large-scale transcriptional profiling of lignified tissues in Tectona grandis. BMC Plant Biol. 2015;15:221.
42. Galeano E, Vasconcelos TS, Ramiro DA, Carrer H. Identification and validation of quantitative real-time reverse transcription PCR reference genes for gene expression analysis in teak (*Tectona grandis* L.f). BMC Res Notes. 2014;7:464.
43. An D, Ma Q, Wang H, Yang J, Zhou W, Zhang P. Cassava C-repeat binding factor 1 gene responds to low temperature and enhances cold tolerance when overexpressed in Arabidopsis and cassava. Plant Mol Biol. 2017;94:109–24.
44. Zhang X, Cheng Z, Zhao K, Yao W, Sun X, Jiang T, et al. Functional characterization of poplar *NAC13* gene in salt tolerance. Plant Sci. 2019;281:1–8.
45. Livak KJ, Schmittgen TD. Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta CT}$ method. Methods. 2001;25:402–8.
46. Kalderon D, Willian DR, Alexander FM, Smith AE. Sequence requirements for nuclear location of simian virus 40 large-T antigen. BMC Res Notes. 1984;7:464.
47. Bent A. Arabidopsis thaliana Floral Dip Transformation Method. Agrobacterium protocols. New Jersey: Humana; 2006. pp. 87–104.
48. Alexander KA, Wakim BT, Doyle GS, Walsh KA, Storm DR. Identification and characterization of the calmodulin-binding domain of neuromodulin, a neurospecific calmodulin-binding protein. J Biol Chem. 1988;263:7544–9.
49. Pant P, Iqbal Z, Pandey BK, Sawant SV. Genome-wide comparative and evolutionary analysis of calmodulin-binding transcription activator (CAMTA) family in Gossypium species. Sci Rep. 2018;8:5573.
50. Lin XZ, Xiao XH, Yang JH, et al. Genome-wide identification and expression analysis of the CAMTA family in rubber tree (*Hevea brasiliensis*). Chin J Trop Crops. 2021;42:2859–68.
51. Zhang N, Chen KL, Bao H, Qi W. Comprehensive analysis of the CAMTA gene family in Gossypium hirsutum. 2022:1–17.
52. Wang G, Zeng H, Hu X, Zhu Y, Chen Y, Shen C, et al. Identification and expression analyses of calmodulin-binding transcription activator genes in soybean. Plant Soil. 2015;386:205–21.
53. Ćmielová J, Řezáčová M. Protein and its function based on a subcellular localization. J Cell Biochem. 2011;112:3502–6.
54. Yang T, Poovaiah BW. A Calmodulin-binding/CGCG box DNA-binding protein family involved in multiple signaling pathways in plants. J Biol Chem. 2002;277:45049–58.
55. Ong C-T, Corces VG. Enhancer function: new insights into the regulation of tissue-specific gene expression. Nat Rev Genet. 2011;12:283–93.
56. Draghici S, Krawetz SA. Global functional profiling of gene expression data. In: Berrar DP, Dubitzky W, Granzow M, editors. A practical Approach to microarray data analysis. Boston: Kluwer Academic; 2003. pp. 306–25.
57. Chinnusamy V, Ohta M, Kanrar S, Lee B, Hong X, Agarwal M, et al. *ICE1*: a regulator of cold-induced transcriptome and freezing tolerance in Arabidopsis. Genes Dev. 2003;17:1043–54.
58. Benn G, Wang C-Q, Hicks DR, Stein J, Guthrie C, Dehesh K. A key general stress response motif is regulated non-uniformly by CAMTA transcription factors. Plant J. 2014;80:82–92.
59. Du L, Ali GS, Simons KA, Hou J, Yang T, Reddy ASN, et al. Ca^{2+}/calmodulin regulates salicylic-acid-mediated plant immunity. Nature. 2009;457:1154–8.
60. Xu Z, Chen M, Li L, Ma Y. Functions and application of the AP2/ERF transcription factor family in crop improvement F. JIPB. 2011;53:570–85.
61. Doherty CJ, Van Buskirk HA, Myers SJ, Thomashow MF. Roles for Arabidopsis CAMTA transcription factors in Cold-regulated gene expression and freezing tolerance. Plant Cell. 2009;21:972–84.
62. Bychkov IA, Kudryakova NV, Kuznetsov VV, et al. Cold stress activates the expression of genes of the Chloroplast transcription Apparatus in Arabidopsis thaliana plants. Dokl Biochem Biophys. 2020;494:235–9.
63. Tomohiro O, Ayuko K, Keita K, Mitsuru K, Masakazu H. F-segments of Arabidopsis dehydrins show cryoprotective activities for lactate dehydrogenase depending on the hydrophobic residues. Phytochemistry. 2020;173:112300.

Publisher's note